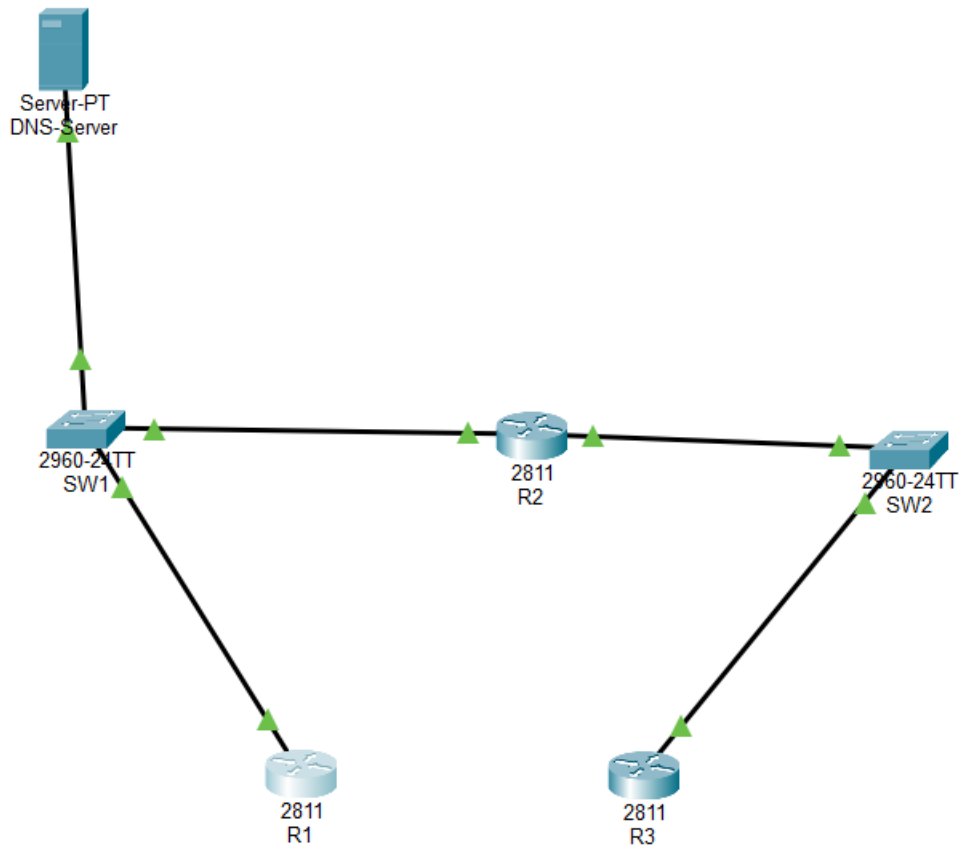


Practical Week 3

Topology



Task 16

```
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#ip domain-lookup
R1(config)#ip name-server 10.10.10.10
R1(config)#ip domain name lab.local
R1(config)#exit
```

Router 1

```
R2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)# ip domain-lookup
R2(config)#ip name-server 10.10.10.10
R2(config)#ip domain-name lab.local
R2(config)#
```

Router 2

```
R3#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#ip domain-lookup
R3(config)#ip name-server 10.10.10.10
^
% Invalid input detected at '^' marker.

R3(config)#ip name-server 10.10.10.10
R3(config)#ip domain-name lab.local
R3(config)#exit
```

Router 3

Task 17

```
R2# ping R1
Translating "R1"...domain server (10.10.10.10)
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.1, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/3 ms
```

Task 18

```
R1#clear arp-cache
R1#ping 10.10.10.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/1/6 ms
```

Task 19

```
R1# show arp
Protocol Address      Age (min)  Hardware Addr  Type   Interface
Internet 10.10.10.1          -    0006.2AE7.A301  ARPA   FastEthernet0/0
Internet 10.10.10.2           1    0090.2193.4A01  ARPA   FastEthernet0/0
Internet 10.10.10.10         1    0060.5C32.5623  ARPA   FastEthernet0/0 Router 1
```

```
R2#show arp
Protocol Address      Age (min)  Hardware Addr  Type   Interface
Internet 10.10.10.1        16    0006.2AE7.A301  ARPA   FastEthernet0/0
Internet 10.10.10.2         -    0090.2193.4A01  ARPA   FastEthernet0/0
Internet 10.10.10.10       33    0060.5C32.5623  ARPA   FastEthernet0/0
Internet 10.10.20.1        17    0090.21C2.0C01  ARPA   FastEthernet0/1
Internet 10.10.20.2         -    0090.2193.4A02  ARPA   FastEthernet0/1 Router 2
```

```
\3>enable
\3#show arp
Protocol Address      Age (min)  Hardware Addr  Type   Interface
Internet 10.10.20.1         -    0090.21C2.0C01  ARPA   FastEthernet0/0
Internet 10.10.20.2        18    0090.2193.4A02  ARPA   FastEthernet0/0 Router 3
```

Task 20

It shows the age of the IP address

Task 21

It does this because ARP is used to resolve the MAC address of the destination device on the network.